ESLAB 2026 · EARLY-CAREER MULTIWAVELENGTH OBSERVATIONS & OUTREACH

From Corona to Community: The EGU LUNEX 2026 Eclipse Campaign in Spain

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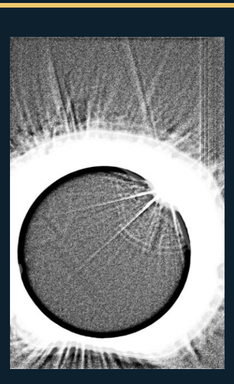
01 Introduction

- Totality reveals the **faint solar corona**, normally hidden by the bright photosphere
- It is still **unclear** how exactly the corona is **heated to 1 MK**, the origins of **slow solar wind**, and the overall **magnetic structures** [5]
- The sudden reduction in solar radiation affects Earth's atmosphere and **radio wave propagation**
- Allows studies of the **ionosphere**
- Provides a unique opportunity for coordinated **multiwavelength observations**

SCIENTIFIC QUESTION

Can complementary, accessible instruments resolve eclipse driven changes in the **corona, sky spectrum and ionosphere**?

02 White light imaging

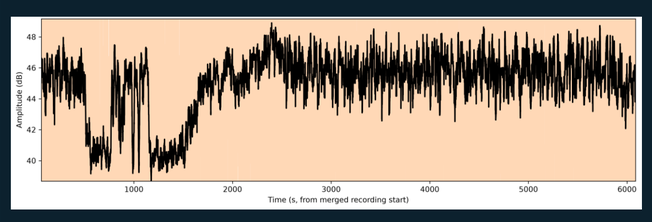


18 FPS · Vaonis Vespera

- **Setup**: Meade LXD75 telescope (15 cm), manual altazimuth mount, CMOS ASI 385 MC Color, focal reducer (0.8x)
- **Exposures**: 5/10/18 FPS, video stacks, dark and flat field corrected
- Overexposed inner corona, well defined **middle corona**
- **Digital Vaonis Vespera:** observations made in Eclipse Mode [9]

03 Radio measurements

KiwiSDR2 · 45 m wire antenna · Station #1 at 23.397 kHz



- Radio observations at the **VLF range**
- **DHO38 station**: German Navy transmitter, **stable signal**
- **Clear 4-5 dB amplitude dip** shortly after the eclipse began
- Signal **unchanged during or after** totality
- The behavior of signal is difficult to predict, it **can increase or decrease** [1,2] due to various reasons (receiver location, interference)
- Our measurements will be compared with other measurements close to Arguedas



ARGUEDAS, SPAIN · 12 AUG 2026 · PROCESSED COLOUR IMAGE

THE CAMPAIGN IN ONE VIEW

Key outcomes

WL	≈4–5 dB	2.8 meters
Recovered middle corona brightness structures	VLF decrease partial phase	GNSS positional shift After totality

PRIMARY INTERPRETATION

- Totality provides **a unique natural experiment** linking the solar corona with the response of Earth's atmosphere and ionosphere
- Coordinated **white light**, **spectroscopic**, **VLF radio** and **GNSS observations** provide complementary measurements across multiple wavelengths
- An **early career** led campaign using **accessible instruments** can deliver valuable data, outreach and preparation for future eclipses

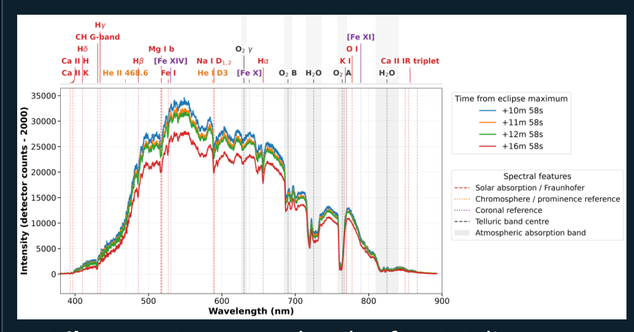
4-5 dB amplitude decrease, 2.8 m displacement and middle corona observations

07 Arguedas eclipse timeline (CEST)



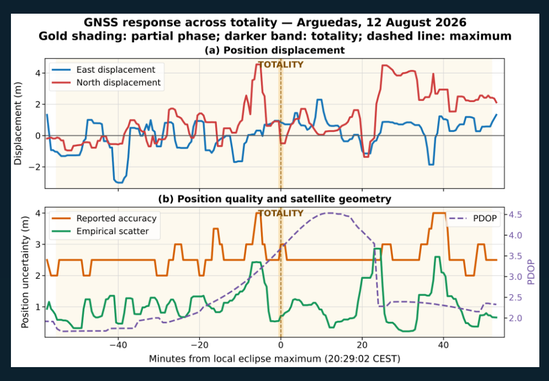
04 Sky spectroscopy

Ocean Optics spectrometer · 400–900 nm



- **Fiber** spectroscopy, shortly after totality
- **Photospheric** light **dominated** spectrum
- Multiple **Fraunhofer absorption features** were identified, including hydrogen Balmer lines, Mg I b, Fe I and Na I D [8]
- **Strong telluric** O_2 and H_2O absorption bands present [8]
- **No coronal** forbidden line emission detected: [Fe XIV], [Fe X] or [Fe XI] [3,4]

05 GNSS experiment



- Single measurement with app ***GNSS logger***
- Clear displacement after ~30min of **+2.8 m**
- Supported by lack of reported decrease from **internal accuracy** and **PDOP**
- Future: **combining data** from all participants

06 Team & logistics



- **8 early career scientists** from 6 countries
- Part of the **official program** in Navarra municipality event in Arguedas (**2000 participants**)
- **Outreach** talks reached nearly **100 listeners**
- Demonstrations, projections, posters available to participants on site

Conclusions & next steps

- **Multi-wavelength observations** during total eclipse allow to better understand **coronal morphology** and **Earth's atmosphere**
- White light imaging is the most insightful about the morphology of corona. Our measurements have an overexposed inner corona, but valuable insight about **the middle corona**
- Radio measurements and GNSS indicate changes in the ionosphere, but the times differ. This is due to the **different layer** responses within the ionosphere [1,2,6]
- Spectroscopy can clearly indicate spectral lines of the photosphere, but it lacks coronal lines due to lack of measurements during totality
- There is large interest from the general public about the unique event of totality, and it is a great place to **cultivate interest in astronomy**
- This was excellent **preparation for the 2nd August 2027 total eclipse in North Africa**, where totality will last up to 6 min 23 s [7]

References · acknowledgements

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- In preparation: Nagainis et al., Multiwavelength observations of the 2026 total solar eclipse from Arguedas.
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